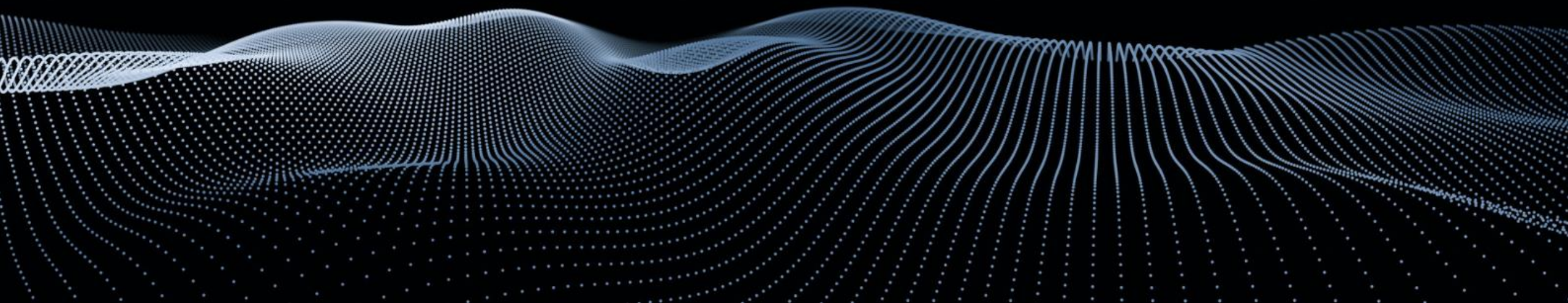


Machine Vision for Deep Fake Detection

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Burke Havranek



Background And Project Goals

- AI generated images and videos are growing more convincing with every image generation model
- Websites, scams, and more are using AI generated images to deceive, misinform, or even extort individuals
- Not many tool are implemented to protect from this manner of misinformation
- A model that can decipher between real and fake photos of people's faces would go far to mitigate these issues
- The model must act fast and give accurate responses about whether the face is real or fake

Dataset

- The data set used was a Kaggle data set comprised of ~130,000 photos of human faces, split almost evenly between real and generated:
 - Real Human Faces: 70,000 256x256 JPEG images (manually verified using random sampling)
 - Fake Human Faces: 63,569 1024x1024 JPEG images (manually verified, differs slightly from author's claim of 64,000 total images)
 - FLUX1 PRO: 3209 Images
 - FLUX1 DEV: 7273 Images
 - SDXL: 53087 Images
 - For this pilot study, we ignore differences in the models used to generate the images



130k Real vs Fake Face
by Shreyansh Patel

Preprocessing

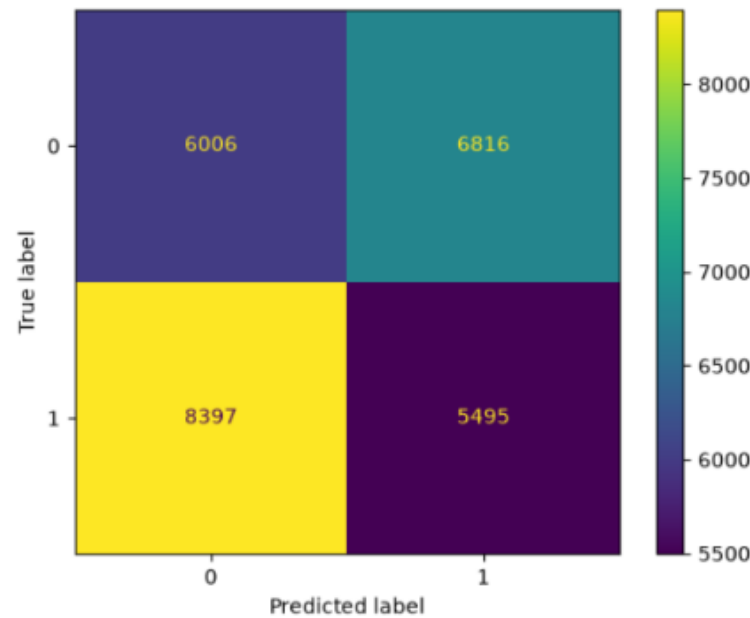
- All generated images were of the same resolution (1024x1024), differing from that of the real images (256x256)
 - To prepare our images for our chosen model, and to remove this inherent bias in the data set, all images were downsampled to 224x224 bitfields using the Lanczos algorithm
- Photographs (real and generated) were taken in a range of different lighting conditions
 - To normalize this data, all images were standardized using the ImageNet standard, and saved as PyTorch tensors
 - Individual color channels were kept so as to not sacrifice valuable information regarding saturation and color grading indicative of AI

Model And Training

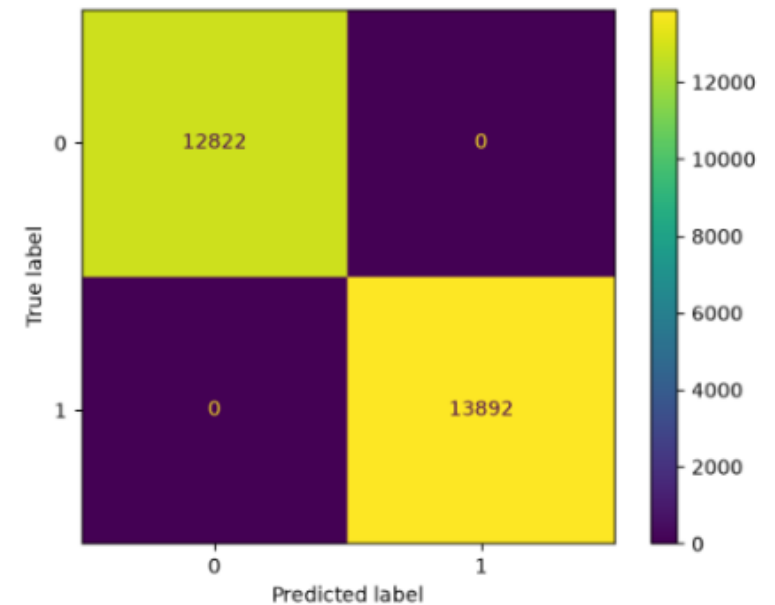
- The model picked was EfficientNet-B0, implemented using PyTorch
- Model produced 1000 output nodes, which were then aggregated using a linear classifier trained at the same time
- Different learning rates were set, $1e-6$ through $1e-3$, with $1e-3$ proving to be the most optimal
 - Hyperparameter tuning was conducted using Adam
- Batch size was set to 64 to optimally utilize the GPUs RAM for our local training with a total epochs of 50 which took 10 hours
- Cross entropy loss was used, and the loss score was the main tool used for evaluating the models while training

Results and Validation

- On epoch 43 and 45, a loss of 0 was recorded for both the training and the verification data sets.
 - Accuracy: 1.000
 - Precision: 1.000
 - Recall: 1.000
 - F1 Score: 1.000



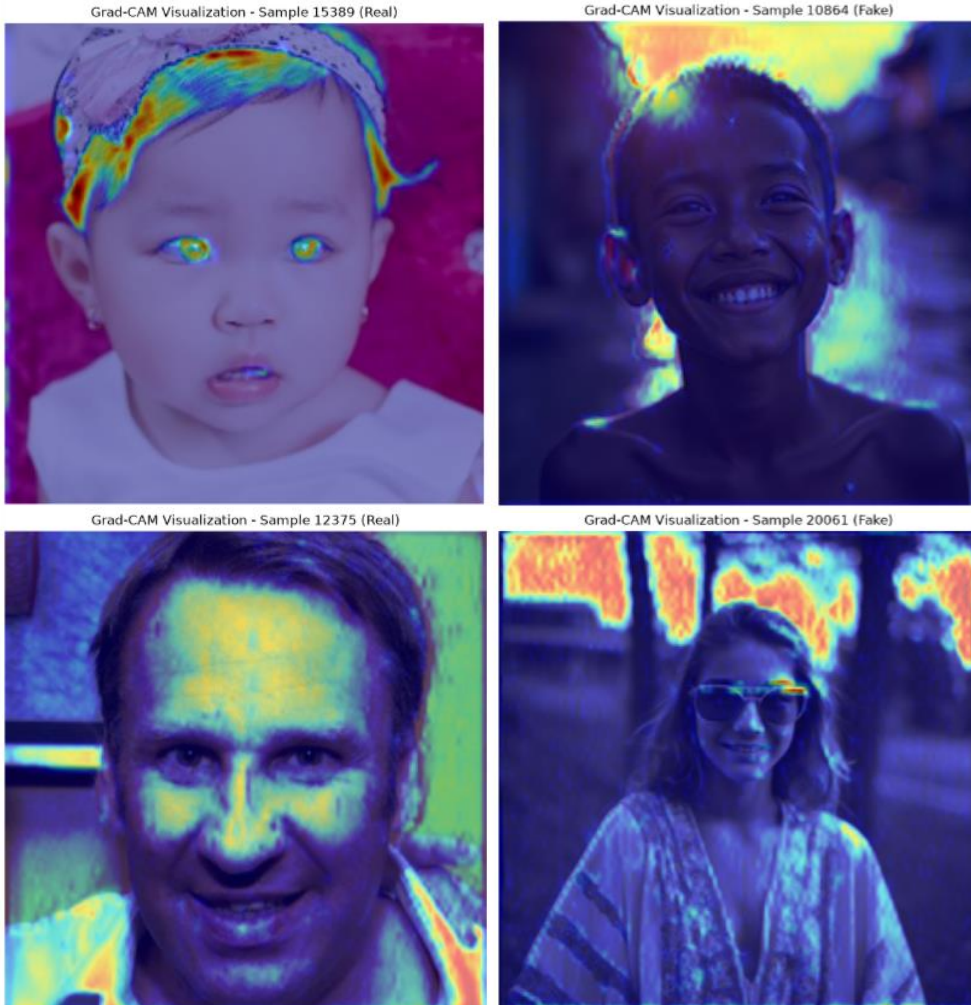
1st Epoch



45th Epoch

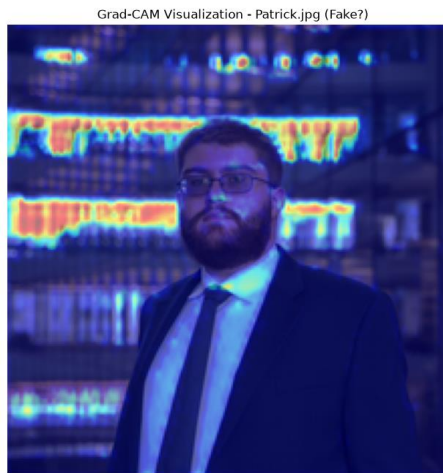
Insights and Visualization

- Using Grad-CAM algorithm, feature can be extracted and visualized



- Eyes, glare, lighting are all features of interest
- Edges of faces with backgrounds are also a source of insight
- Appears to confirm initial suspicions of high-contrast and -saturation images being indicative of AI generation

Unseen Data



- High confidence in wrong classifications in data following the same encoding method points toward significant overfitting, potentially requiring significantly more data to overcome



Future Work

- Regress model to previous iterations to see if the overfitting issue might not be solvable; to that end, split validation data into training and unseen data
- Add subcategories of other classification (model type, age, etc.)
- Train the model with more data and on a larger scale, perhaps taking into account computing services like AWS
- Create integration with either a drop GUI or a web plug-in using this model, which would automatically highlight AI images